

**An Econometric Analysis of UnitedHealth Group's Stock Returns Using the
Fama-French Five-Factor Model**

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Executive Summary

This paper asks whether the Fama-French risk factors explain UnitedHealth Group's daily excess stock returns. The analysis uses UNH daily returns from Bloomberg historical prices and daily Fama-French factors from 2021-05-03 to 2026-03-31, for 1,234 merged observations. CAPM, FF3, and FF5 models are estimated by ordinary least squares. The main finding is that the market factor is positive and statistically significant in every specification, while UNH's estimated beta remains below one. Alpha is not statistically significant, so the results do not support abnormal excess returns after factor controls. The final FF5 model has an R-squared of about 0.066, meaning the model explains only a modest share of daily excess-return variation. Diagnostics do not detect heteroskedasticity using the Breusch-Pagan and White tests, but residuals are non-normal and fat-tailed. Overall, systematic factors matter, but firm-specific shocks remain important for UNH's daily returns.

Abstract

This paper examines whether the Fama-French risk factors explain UnitedHealth Group's daily excess stock returns. The analysis uses Bloomberg PX_LAST historical price data for UNH and Kenneth French daily five-factor data. The final merged sample covers 2021-05-03 to 2026-03-31 and includes 1,234 observations. Returns and factors are measured in percent, and the dependent variable is UNH_Excess_Return. Three OLS specifications are estimated: CAPM, the Fama-French three-factor model, and the Fama-French five-factor model. Across the specifications, the market coefficient is positive and statistically significant at the 1 percent level, with beta estimates ranging from about 0.41 to 0.51. This indicates that UNH comoves with the market but has below-one market sensitivity. Alpha is approximately -0.04 and is not statistically significant at conventional levels. The FF5 model has an R-squared of 0.0659, which indicates limited explanatory power for daily single-stock returns. The Breusch-Pagan and White tests fail to reject homoskedasticity, while the Anderson-Darling test rejects normality. Robust HC1 standard errors are reported as a sensitivity check and do not change the main conclusion about the market factor.

1. Introduction

This paper asks whether the Fama-French risk factors explain UnitedHealth Group's daily excess stock returns. The question matters because factor models are designed to separate systematic market exposure from firm-specific movements. For a single stock such as UNH, this distinction is important. A company can have low estimated market sensitivity and still experience large daily return shocks if firm-specific information is important.

UnitedHealth Group is a useful case for this question because it is a large health-care and managed-care company with substantial business scale, broad operating exposure, and visible stock-price movement during the sample. The Bloomberg DES screen classifies UNH as Managed Care and reports a beta versus the S&P 500 Index of 0.61 as May 2026 support context. That market screen is not an input to the regression, but it makes the empirical question natural: if UNH appears low beta in market screens, do the daily return regressions also show below-market sensitivity?

The paper answers the question using daily Bloomberg PX_LAST prices and daily Fama-French factors. The data are merged by date, daily returns are computed from closing prices, and the dependent variable is UNH excess return. Three models are estimated: CAPM, FF3, and FF5. This structure allows the paper to compare a market-only model with specifications that add size, value, profitability, and investment factors. The class OLS framework and the local R econometrics text provide the basic estimation and diagnostic setting for this analysis.

The main finding is that the market factor is the strongest and most reliable predictor. It is positive and highly significant in all three models, and its coefficient remains below one. Alpha is not statistically significant in any model. The non-market factors add some joint explanatory

power, but the R-squared values remain low. This means that broad factor exposures explain only a modest portion of UNH's daily return variation.

2. Company Background and Model Framework

UnitedHealth Group Incorporated is the company analyzed in this paper. The local Form 10-Q identifies UnitedHealth Group Incorporated as the registrant and lists UNH common stock on the New York Stock Exchange. The local Bloomberg DES screen classifies the company as Managed Care, lists the headquarters in Eden Prairie, Minnesota, and reports about 390,000 employees. The same screen reports market capitalization of about \$364.0 billion, about 908.1 million shares outstanding, a 52-week high of \$404.15 on 05/13/26, and a 52-week low of \$234.60 on 08/01/25. These Bloomberg values are used only as descriptive support context, not as regression inputs.

The Bloomberg DES and dividend screens also show a mature-company profile. The DES screen reports a 12-month total return of 33.84 percent, beta versus SPX of 0.61, indicated gross dividend yield of 2.21 percent, five-year net dividend growth of 12.07 percent, trailing 12-month EPS of 15.50, and estimated EPS of 18.36. The dividend screen also reports a 12-month yield of 2.21 percent and five-year dividend growth of 12.07 percent. These facts support the company background, but the paper's statistical evidence comes from the daily return regressions.

The Bloomberg 2025 revenue breakdown allocates about \$312.730 billion to UnitedHealthcare and \$130.917 billion to Optum out of \$447.567 billion in consolidated revenue, with a separate adjustment category. These displayed allocations are roughly 70 percent and 29 percent of the total. They describe the screen's consolidated revenue allocation, rather than gross segment revenues before eliminations. The business mix provides context for why a broad factor model may leave healthcare-specific and company-specific movements unexplained.

The Bloomberg financial statement files show that revenue rose from \$400.278 billion in 2024 to \$447.567 billion in 2025, while profitability weakened. In the adjusted income-statement view, 2025 EBITDA was about \$27.474 billion, adjusted net income was about \$14.142 billion, adjusted EPS was about \$15.52, cash from operations was about \$19.697 billion, and free cash flow was about \$16.075 billion. Current or LTM cash from operations was about \$23.153 billion, and current or LTM free cash flow was about \$19.666 billion. At year-end 2025, cash, cash equivalents, and short-term investments totaled about \$28.121 billion and total debt was about \$83.004 billion. The profitability file reports ROE of 12.91 percent, ROA of 3.97 percent, EBITDA margin of 5.57 percent, net income margin of 2.69 percent, and a dividend payout ratio of 65.75 in 2025. Cash and cash equivalents alone were \$24.365 billion. The adjusted income-statement view and the ratio screen use different measures: the reported 5.57 percent EBITDA margin is from the ratio screen and is not the margin implied by the adjusted income-statement EBITDA figure.

The Form 10-Q covering the quarter ended March 31, 2026 provides retrospective context for the sample endpoint. Q1 2026 revenues were \$111.721 billion versus \$109.575 billion in Q1 2025. Earnings from operations were \$8.990 billion versus \$9.119 billion, and net earnings attributable to common shareholders were \$6.280 billion versus \$6.292 billion. Cash from operating activities was \$8.912 billion versus \$5.456 billion, cash and cash equivalents were \$28.001 billion, long-term debt was \$71.440 billion, common dividends paid were \$2.005 billion or \$2.21 per share, and shares outstanding as of April 30, 2026 were 908,144,404. Management discussion states that revenue growth was driven by pricing trends at UnitedHealthcare and growth at Optum Rx, partly offset by fewer people served in Medicare Advantage, commercial risk-based offerings, and Medicaid, along with fewer patients served under value-based

arrangements at Optum Health. The filing also reports UnitedHealthcare operating earnings up 9 percent and Optum operating earnings down 15 percent. The filing was available after the sample endpoint and is not treated as an explanatory variable observed throughout the sample. The factor-model framework connects this company context to the return data. The CAPM model is $UNH_Excess_Return_t = \alpha + \beta_1 Mkt_RF_t + u_t$. The FF5 model is $UNH_Excess_Return_t = \alpha + \beta_1 Mkt_RF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + u_t$. UNH_Excess_Return is the UNH daily return minus RF. Mkt_RF is the market excess return, SMB is Small Minus Big, HML is High Minus Low, RMW is Robust Minus Weak, CMA is Conservative Minus Aggressive, RF is the risk-free rate, and u_t is the error term.

3. Data and Descriptive Statistics

The empirical analysis uses daily data. UNH prices come from the Bloomberg historical-price file using PX_LAST . The Fama-French factors come from the daily five-factor file. RF also comes from the Fama-French file, not from a Bloomberg Treasury screen. The first return observation is lost because return construction requires the previous trading day's price.

The stock return is computed as $UNH_Return_t = ((PX_LAST_t / PX_LAST_(t-1)) - 1) * 100$.

The excess return is computed as $UNH_Excess_Return_t = UNH_Return_t - RF_t$. Returns and factors are kept in percent, so the regression coefficients have a percentage-point interpretation.

The final sample uses only overlapping dates after merging the UNH return file and the Fama-French factor file by date. The merged sample runs from 2021-05-03 to 2026-03-31 and contains 1,234 observations. This is a price-derived return measure; the baseline does not separately add cash dividends or implement dividend reinvestment.

No valid observations were removed, trimmed, or winsorized. Extreme observations are kept because they reflect actual stock return behavior unless there is clear evidence of data error. The paper handles outliers through diagnostic testing and robust standard errors rather than deleting valid market movements.

Table 1. Descriptive Statistics

Variable	Obs.	Mean	Median	Std. Dev.	Min.	Max.
UNH_Excess_Return	1,234	-0.024925136	0.03529529	1.980748818	-22.39967	11.95834
Mkt_RF	1,234	0.030340357	0.04000000	1.115246348	-5.92000	9.65000
SMB	1,234	-0.019781199	-0.04000000	0.694038828	-2.53000	4.20000
HML	1,234	0.024489465	-0.00500000	0.907521928	-3.89000	3.64000
RMW	1,234	0.017228525	0.00000000	0.659952271	-2.23000	4.25000
CMA	1,234	0.007074554	0.00000000	0.589601663	-2.92000	2.37000
RF	1,234	0.013881686	0.02000000	0.008642992	0.00000	0.02000

Table 1 shows that the mean daily UNH excess return is close to zero and slightly negative, while the median is slightly positive. The standard deviation of 1.980748818 percent is much larger than the mean, which is typical for daily stock-return data. The minimum of -22.39967 percent and maximum of 11.95834 percent show that the series contains large daily movements. This supports caution when interpreting daily return regressions because the average return is small relative to daily volatility.

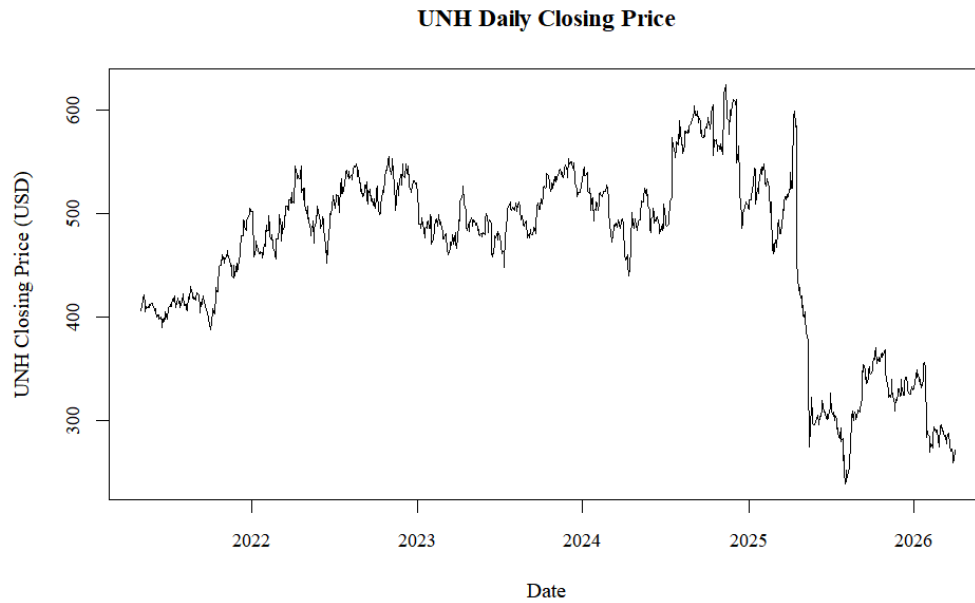


Figure 1. UNH Daily Closing Price, May 2021 to March 2026.

Figure 1 shows the UNH daily closing price over the regression sample. The stock traded at relatively high levels through much of 2024 before a sharp decline in 2025. This price path reinforces the importance of firm-specific shocks.

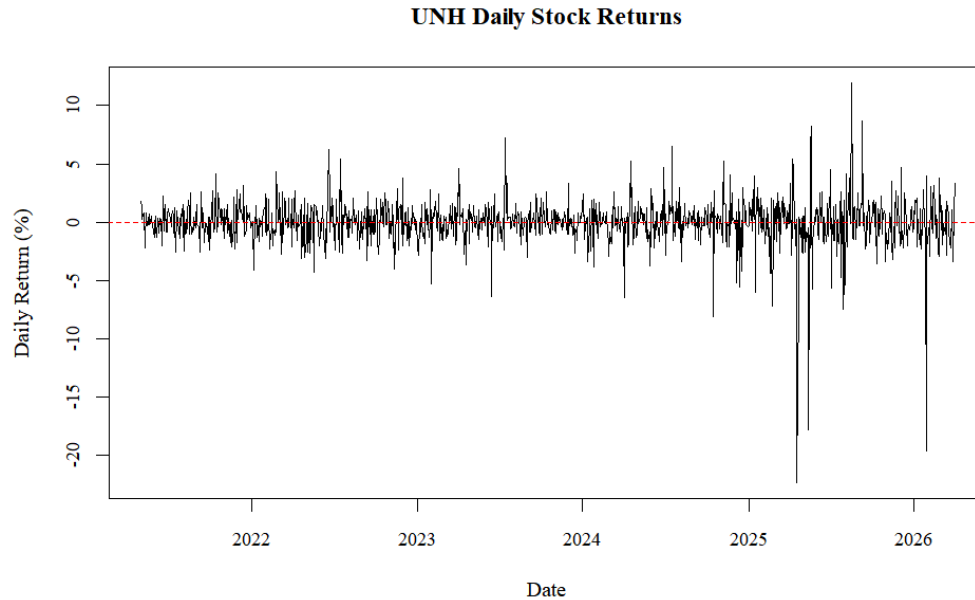


Figure 2. UNH Daily Stock Returns.

Figure 2 shows that daily returns are centered near zero but include large positive and negative observations. These movements help explain why the factor model can be statistically significant while still having low explanatory power.

4. Econometric Model

The paper estimates three OLS specifications. The CAPM model is $UNH_Excess_Return_t = \alpha + \beta_1 Mkt_RF_t + u_t$. The FF3 model is $UNH_Excess_Return_t = \alpha + \beta_1 Mkt_RF_t + \beta_2 SMB_t + \beta_3 HML_t + u_t$. The FF5 model is $UNH_Excess_Return_t = \alpha + \beta_1 Mkt_RF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + u_t$.

Because returns and factors are measured in percent, each coefficient measures the predicted percentage-point change in UNH excess return for a one percentage-point change in the factor, holding other included factors constant. Alpha measures the average excess return not explained by the included factors. If alpha is statistically insignificant, the model does not provide evidence of abnormal excess return after controlling for factor exposures.

The OLS assumptions used in the class framework are that the model is linear in parameters, there is no perfect multicollinearity, the error has zero conditional mean, the error variance is constant for usual OLS standard errors, and normality supports exact small-sample inference. Because the sample has 1,234 observations, large-sample interpretation is also relevant. The paper reports conventional standard errors and then checks robust HC1 standard errors as a sensitivity test. These diagnostics do not establish zero conditional mean. Large-sample inference still requires appropriate moment and dependence conditions; a large observation count alone does not ensure valid inference.

5. Empirical Results

Table 2 reports CAPM, FF3, and FF5 side by side. Entries labeled "Not included" indicate that a factor is not included in that model. Standard errors are reported in parentheses, and the significance stars follow the class regression output.

Table 2. Regression Results

Variable	CAPM	FF3	FF5
Alpha	-0.037 (0.055)	-0.048 (0.055)	-0.048 (0.055)
Mkt_RF	0.406*** (0.049)	0.488*** (0.055)	0.512*** (0.056)
SMB	Not included	-0.126 (0.084)	-0.078 (0.094)
HML	Not included	0.230*** (0.067)	0.106 (0.086)
RMW	Not included	Not included	0.077 (0.104)
CMA	Not included	Not included	0.269** (0.120)
Observations	1,234	1,234	1,234
R-squared	0.052	0.062	0.066
Adjusted R-squared	0.052	0.059	0.062

*Note: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

In the CAPM regression, the Mkt_RF coefficient is 0.40607 with a p-value of 4.24e-16. It is positive and statistically significant at the 1 percent level. A one-percentage-point increase in market excess return is associated with about a 0.406 percentage-point increase in UNH excess return. Because the coefficient is below one, UNH is less sensitive to the market than a beta-one stock in this specification. Alpha is -0.03725 with a p-value of 0.498, so it is not statistically significant.

In the FF3 regression, the Mkt_RF coefficient remains positive and significant at 0.48799 with a p-value below 2e-16. SMB is -0.12630 with a p-value of 0.132817, so it is not statistically significant. HML is 0.22996 with a p-value of 0.000598, so it is significant in the three-factor model. Alpha remains insignificant, with p-value 0.382740. The R-squared rises from 0.05227 in CAPM to 0.06157 in FF3, which is an improvement but still modest.

In the FF5 regression, Mkt_RF is 0.51161 with a p-value below $2e-16$. This is again positive, highly significant, and below one. SMB, HML, and RMW are not statistically significant in the full model, with p-values of 0.4077, 0.2158, and 0.4602. CMA is 0.26881 with a p-value of 0.0249 under conventional standard errors, so it is significant at the 5 percent level in the main table. Alpha is -0.04781 with a p-value of 0.3825, so the FF5 model also does not show significant abnormal excess return.

The model comparison supports a careful interpretation. The non-market factors jointly improve the specification relative to CAPM, with p-value 0.001348 for the joint test that SMB, HML, RMW, and CMA equal zero. However, the added FF5 pair RMW and CMA is only marginal beyond FF3, with p-value 0.05831, and the FF3 to FF5 incremental test has p-value 0.058308. The model is useful, but it is incomplete for single-stock daily returns.

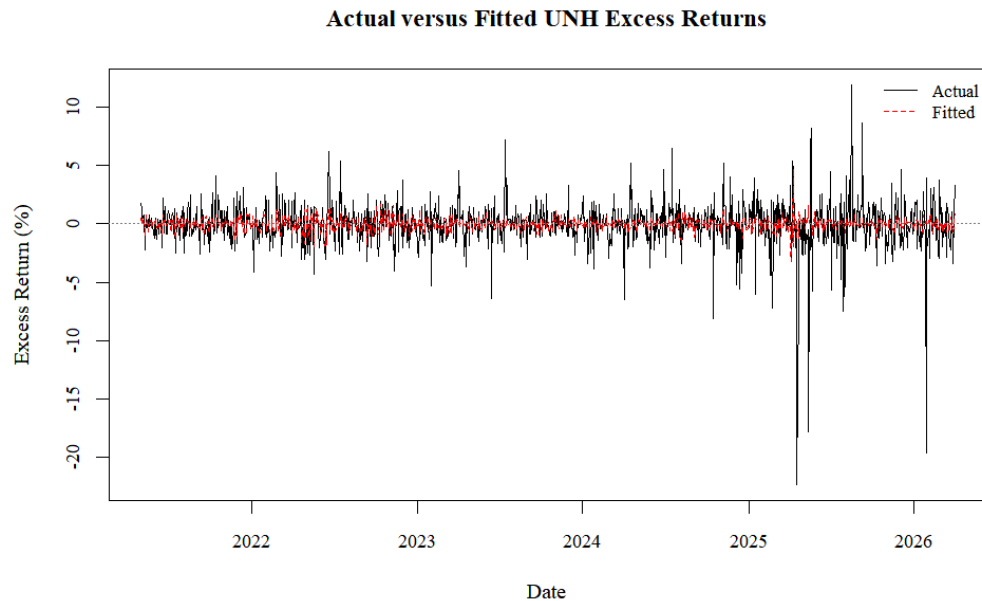


Figure 3. Actual versus Fitted UNH Excess Returns from the Fama-French Five-Factor Model.

Figure 3 compares actual UNH excess returns with fitted values from the FF5 model. The fitted values are much smoother than actual returns. This supports the low R-squared interpretation and shows that the model captures some systematic factor movement but does not explain large firm-specific daily shocks.

6. Diagnostic Testing and Robustness Checks

Table 3 summarizes the diagnostic and robustness checks. The Breusch-Pagan test is the primary heteroskedasticity test in the body. The null hypothesis is that the error variance is constant, meaning homoskedasticity. The decision rule is to reject homoskedasticity if the p-value is below 0.05 and fail to reject if the p-value is at least 0.05.

Table 3. Diagnostic Tests and Robustness Checks

Check	Statistic or result	p-value	Conclusion
VIF	Mkt_RF 1.302836; SMB 1.439641; HML 2.018858; RMW 1.579036; CMA 1.670055	n/a	All below 5
Breusch-Pagan	BP = 2.3722, df = 5	0.7956	Fail to reject homoskedasticity
White test	White LM = 0.03363219	0.9833245	Same conclusion as BP
Durbin-Watson	DW = 1.915	0.06759	No 5 percent rejection
Anderson-Darling	A = 38.353	< 2.2e-16	Reject normality
Robust HC1	CMA robust p-value = 0.05454	0.05454	CMA becomes marginal

The Breusch-Pagan p-value is 0.7956, so the paper fails to reject homoskedasticity at the 5 percent level. The White test p-value is 0.9833245 and leads to the same conclusion. The formal tests do not provide evidence of heteroskedasticity. The residual plots show outliers, but outliers and non-normality are not the same thing as heteroskedasticity. Here Breusch-Pagan is the studentized test from `lmtest`. The White calculation uses squared residuals regressed on fitted values and their square, with two degrees of freedom; it is not the full regressor-interaction version.

Robust HC1 standard errors are included as a sensitivity check, not because heteroskedasticity was formally proven. Under robust HC1 standard errors, Mkt_RF remains highly significant, alpha remains insignificant, and SMB, HML, and RMW remain insignificant. CMA changes from significant under conventional standard errors to marginal under robust standard errors,

with robust p-value 0.05454. The main conclusion about the market factor does not change. HC1 does not correct serial correlation or reduce the influence of extreme observations on the OLS coefficients.

The VIF values are all below 5, so multicollinearity does not appear to be a major issue. Against the alternative of positive serial correlation, the Durbin-Watson p-value is 0.06759, so the test does not reject no serial correlation at the 5 percent level, although weak serial correlation may be a limitation at a less strict level. The Anderson-Darling test strongly rejects normality with p-value below $2.2e-16$. The Q-Q plot in the appendix shows tail deviations consistent with fat-tailed residuals.

The daily return series contains extreme observations. These observations may contribute to non-normal residuals and could raise visual concern in residual plots. The baseline specification retains all valid trading-day observations. Extreme return days are treated as economically meaningful events rather than data errors, so the paper does not mechanically trim the sample.

7. Discussion

The results support a below-one market exposure interpretation. The market coefficient is positive and significant across all models, but it remains below one. This means UNH comoves with the market, but the estimated sensitivity is lower than the market portfolio's one-for-one movement. The result is consistent with the Bloomberg beta screen, but the regression result is the paper's main evidence.

The peer screens add useful context. UNH traded at a premium to peer medians on P/E, forward P/E, and P/FCF. Its one-year total return was much stronger than the peer median, and its adjusted beta was near 0.61. At the same time, UNH's volatility standard deviation was above the peer median. This distinction matters because low market beta does not mean low total volatility.

It fits the low R-squared result because broad market exposure is limited, but total daily variation can still be large.

The earnings file helps explain some of the large residual behavior. Major return days line up with earnings announcements. On 04/17/2025, Q1 2025 reported EPS was 6.863 versus an estimate of 7.274, and the stock price change was -22.38 percent. On 07/29/2025, Q2 2025 reported EPS was 3.776 versus an estimate of 4.589, and the price change was -7.46 percent. On 01/27/2026, Q4 2025 reported EPS was 2.141 versus an estimate of 2.102, and the price change was -19.61 percent. These observations show that some extreme daily returns are real company-event days, not bad data. The workbook distinguishes reported EPS from comparable EPS, and its surprise percentage uses the comparable field. These date matches are descriptive: the negative January return despite reported EPS above the listed estimate shows why an EPS comparison alone cannot identify the cause of a stock-price change.

The low R-squared values therefore have a clear economic interpretation. The FF5 model captures part of systematic factor movement, especially market exposure, but much of UNH's daily variation remains unexplained. Firm-specific information and omitted short-run shocks are possible contributors; the regression does not identify each residual's cause. The paper does not directly model earnings surprises, regulatory announcements, health-care policy events, litigation news, or analyst revisions. Future work could add those variables to test whether they explain the large residual days.

8. Conclusion

This paper tests whether the Fama-French risk factors explain UnitedHealth Group's daily excess stock returns from 2021-05-03 to 2026-03-31. The main finding is that the market factor is positive and highly significant in all specifications. The market coefficient is below one in

CAPM, FF3, and FF5, which supports the interpretation that UNH had below-market sensitivity during the sample.

The paper does not find significant alpha. The intercept is statistically insignificant in every model, so the results do not support abnormal excess returns after controlling for the included factors. The non-market factors add some joint explanatory power, but individual factor evidence is limited in the FF5 model. SMB, HML, and RMW are not significant in FF5. CMA is significant under conventional standard errors but becomes only marginal under robust HC1 standard errors.

The diagnostic results also support a cautious conclusion. The Breusch-Pagan and White tests do not show strong evidence of heteroskedasticity, and robust standard errors do not overturn the main result. VIF values are below 5. Residual normality is rejected, and the Durbin-Watson test suggests weak possible serial correlation at a less strict level but not a 5 percent rejection.

Overall, the FF5 model is useful for measuring systematic exposure, but it explains only a modest share of UNH's daily excess-return variation.

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Appendix

Appendix Table A1. Bloomberg Company Snapshot and Operating Context

Item	Value	Local source
Classification	Managed Care	Bloomberg DES
Headquarters	Eden Prairie, Minnesota	Bloomberg DES
Employees	About 390,000	Bloomberg DES
Market cap	About \$364.0B	Bloomberg DES
Consolidated revenue allocation	UnitedHealthcare about 70%; Optum about 29%	modl.gif
Profitability context	ROE 12.91%; ROA 3.97%; EBITDA margin 5.57%; net income margin 2.69%	ratiosFOFAUNH.pdf
Q1 2026 context	Revenues \$111.721B; operating earnings \$8.990B; operating cash flow \$8.912B	10-Q

Appendix Table A2. Bloomberg Peer Valuation and Market Context

Context item	UNH	Peer comparison
P/E	25.86	15.51 peer median
Forward P/E FY1	21.83	14.76 peer median
P/FCF	18.49	13.66 peer median
One-year total return	About 33.66% to 33.84%	About 8.30% peer median
Adjusted beta	0.61	0.61 peer median
Volatility standard deviation	5.10	3.02 peer median

Appendix Table A3. FF5 Robust HC1 Standard Error Comparison

Variable	Estimate	OLS p-value	Robust HC1 SE	Robust p-value
Alpha	-0.047814	0.3825	0.054388	0.37950
Mkt_RF	0.511605	< 2e-16	0.055060	< 2e-16
SMB	-0.078220	0.4077	0.097863	0.42428
HML	0.105915	0.2158	0.092333	0.25157
RMW	0.076838	0.4602	0.096554	0.42630
CMA	0.268808	0.0249	0.139686	0.05454

The accompanying repository contains run.R and numbered scripts in R/. The scripts load the Bloomberg price workbook and saved Kenneth French daily factors, compute excess returns, estimate CAPM, FF3, and FF5, and run the diagnostics reported here. The repository also records the data requirements and reproduction checks.

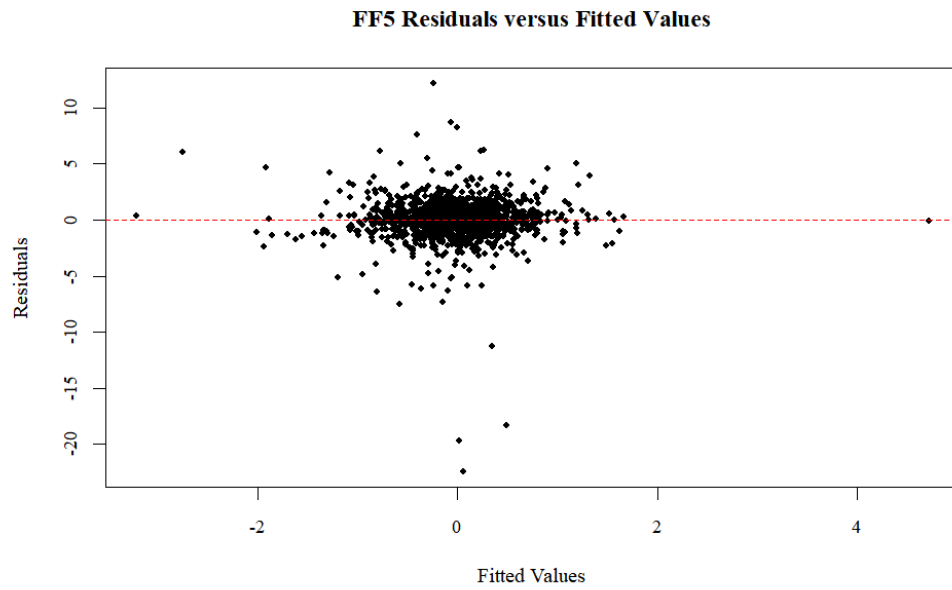


Figure A1. FF5 Residuals versus Fitted Values.

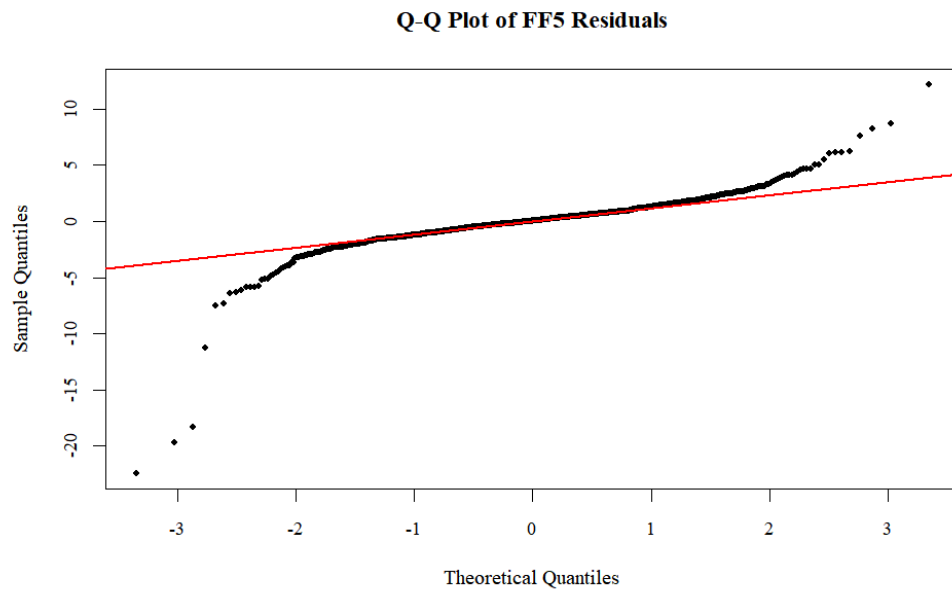


Figure A2. Q-Q Plot of FF5 Residuals.